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INTRODUCTION TO DATABASES

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Entities and entity instances

- **Entity** is anything that the data could be collected of
  - People, Student, Cat, Animal, Course, Invoice, Invoice item, Payment method, etc.

- **Entity instances** are exact objects from the real world
  - One specific **John** is an **instance** of the **People** entity
  - **AMEX** credit card is an **instance** of the **Payment methods** entity
  - **The fourth semester** is an **instance** of the **Semesters** entity

- Entities can be observed as structures, instances can be observed as objects

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Attributes

- Entities contains describing attributes
  - Entity **People** contains **Day of birth** attribute
  - Entity **Cat** contains **Fur color** attribute

- **Each entity instance contains its' attribute values**
  - **Instance** John **contains value** 7. 3. 2022. for the Day of birth **attribute**
  - **Instance** Garfield **contains value** Orange for the Fur color **attribute**

- A value is added to an attribute of entity

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Attribute dependencies

- Attribute values may depend to some parameter, ie. time
  - An instance of People entity have attribute Age value: 9125 days. Tommorow it will be 9126, the day after tommorow 9127 etc.

- Those kind of attributes should be avoided in databases
  - **Day of birth** attribute is more convenient than **Age** attribute
  - Age could be calculated

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Data and information

- Elementary information is defined with entity instance, attribute and its' value
- Example: "Join is 185 cm tall"
    - Entity: People
    - Entity instance: John
    - Attribute: Height
    - Attribute value: 185 cm
- Data is the written information
    - Written on the paper
    - In computer memory
    - On disk,...

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Databases

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DBMS and databases

- Database is a set of connected data

- Database management system is called DBMS (DataBase Management System)
  - It contains one or more databases
  - It is used by users or applications when working with databases
  - It handles database data storage
  - It provides a set of services, like security services, administrative services etc.

- If the DBMS is an entity, then SQL Server, PostgreSQL, Oracle, mySQL and DB/2 are the instances of that entity

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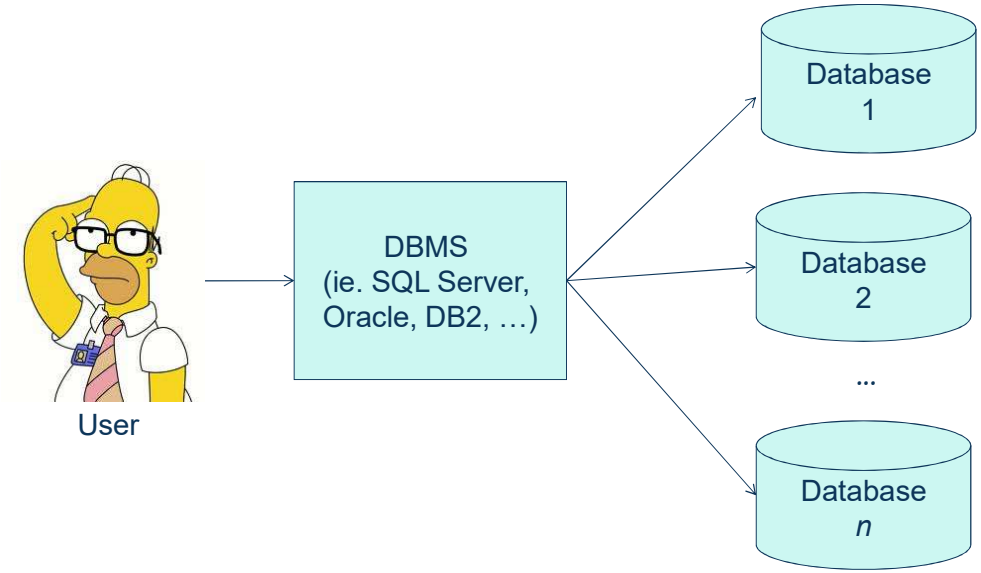
DBMS and databases



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Data models

- Data model is a set of rules that defines how database logical structure looks like

- Data models:
  - Network model
  - Hierarchical data model
  - **Relational data model**
  - Object oriented data model
  - Spatio-temporal data model

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Object oriented data model

- Data stored as an object, like in object-oriented programming languages
- Disadvantages:
  - There are no flexible query languages
  - Bad connection with standard reporting systems, OLAP, connecting with other databases,...
  - Lack of formal mathematical base
- Examples: db4o, ORION, DTS/S1, Perst, ...

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Relational data model

- The most common model nowadays
- The slowest model, but most flexible
- Based on mathematical term relation (table)
  - Data is shown in tables (table model)
  - Example: table City

IDCity	CityName	NumberOfResidents	CountyID
1	Zagreb	792875	1
2	Split	178192	4
3	Rijeka	128735	3
4	Osijek	107784	6
5	Dubrovnik	38474	4

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Relational database management systems

- Relational DBMSs are called RDBMSs (Relational DataBase Management Systems)
- The most popular RDBMSs:
  - Oracle – from 1979. Oracle v2
  - Microsoft SQL Server
    - from 1989., taken over from Sybase based on OS/2
    - from 1993., based on Windows NT, SQL Server 4.21
  - IBM DB2 – from 1987.
  - PostgreSQL – University of California, Berkeley, from 1989.
  - MySQL – Sweden, MySQL AB, from 1995.

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History of relational model

- The theoretical basis created by **Edgar Frank Codd** (19 August 1923 – 18 April 2003) in the 1960s
- Initial implementations were slow and inefficient
  - The relational model only implicitly (using attributes) indicates relationships between entity instances
  - Problem: It takes time to establish connections when querying
  - Advantage: user-defined connections!
  - Sacrificed little speed for flexibility
- The increase in computing power has enabled the development of commercial RDBMS
- It has been the predominant model since the mid-1980s

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Database life cycle

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Database life cycle

- Database life cycle starts with an idea of a database, and it ends if the database is no longer used
- Database life cycle may consist of:
  - The analysis of a needs
  - Data modeling
  - Implementation
  - Testing
  - Maintenance
- Database lifecycle is often tied to the application life cycle

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Relations between the entities

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Relations between the entities (1/2)

- Entities can be in a certain relation with each other
  - We name relations (Works, Belongs, Drives, Writes, ...)
  - For example, "Student enrolled in a course“
- A relation is established between two or more instances of an entity
  - For example, “John Doe enrolled in ItD“
- Each instance can have an unlimited number of relations with other instances of different entities
  - For example,
    “A worker John loads truck“, “A worker Marko is employed by BringItAndRun l.l.c."

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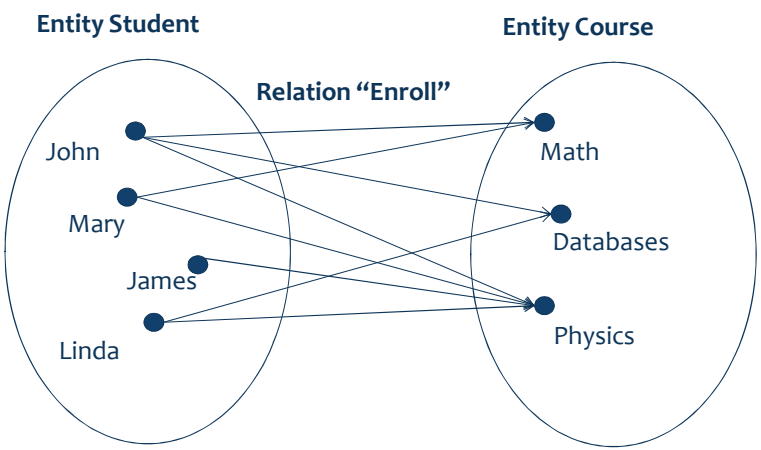
Relations between the entities (2/2)

- Relations between entities can be represented as a mappings set



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Relation classification (1/3)

- We can classify relations between entities in various ways
- Of particular importance is the classification according to the number of instances of entities that can participate in the relation:

- Relation **One-to-one (1: 1)**
  - One instance from the first set can be mapped to a maximum of one instance from the second set
  - The reverse is also true
  - Examples:
    - Relation **Belonging** between the entities of the **State** and the **Anthem**
    - Relation **Marriage** between **Husband** and **Wife**
    - Relation between John and his SSN (we can also see this as the relation between the Citizens and the SSN)

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Relation classification (2/3)

- Relation **One-to-many (1: N)**
  - One instance from the first set can be mapped to any number of instances from the second set
  - The reverse is not true: one instance from the second set can be mapped to a maximum of one instance from the first set
  - For example:
    - Relation **Belongs** between the entities **State** and **City**
    - Relation **Lecturer** between the entities **MainTeacher** and **Course**
    - Relation **Belongs** between the entities **Invoice** and **InvoiceItem**

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Relation classification (3/3)

- Relation **Many-to-many (M: N)**
  - One instance from the first set can be mapped to any number of instances from the second set
  - The reverse is also true
  - For example:
    - Relation **Enrolled** between **Student** and **Course**
    - Relations 1: 1, N: 1 and 1: N are special cases of the M: N relation
    - Examples: The attitude of specialist doctors towards patients

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Key attributes (1/2)

- How to distinguish two instances within an entity?
- The values of one or more attributes uniquely determine the entity instance - key attribute(s)
  - Entity Account has the KeyAccountNumber attribute
  - Entity Diagnosis has the key attribute consisting of DoctorSSN and PatientSSN

- Each entity can have multiple attributes that make unique value
  - Entity Person has SSN, Fingerprint, DNA code,…
  - Do the combination of the attributes Name, Surname, Father's name, Mother's name, Date of birth and Place of birth make the key attributes?

- For each entity, we must select one or more key attributes in order to distinguish its instances
  - Preferably, the values of the key attribute do not change

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Key attributes (2/2)

- Choosing a key attribute is often not trivial
- What is the key attribute for the entity Person, if we also consider persons who are from different countries? SSN is not a candidate
- What is the key attribute for the InvoiceItem entity?
- What is the key attribute for the Vehicle entity?

- Best practice is to assign a surrogate attribute to each entity
  - A surrogate attribute does not exist in the real world but has its meaning only in the context of a database
- The surrogate attribute will be denoted by EntityName
  - For example, for the Vehicle entity, we will have the IDVehicle attribute

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ER MODEL

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Modelling database

- The fundamental question: how to get a database from user requests?
- Equivalent to the question of how to design a relational database schema
  - We will learn relational schemes later
- Design process:
  - For the most part, it is not formally defined
  - It is subjective and requires creativity
  - It gets easier with experience

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ER model

- The ER model is an abstraction of the real world
- The ER model represents the intermediate step between user requests and the database (i.e. relation schema or relationship model)
- The ER model is much more precise than user requirements but less accurate than the relationship model

- An issue:
  - There are well-defined ways to transform an ER model into a relationship model
  - There are no well-defined ways to transform user requirements into an ER model
- Most of the work is to make a good ER model

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ER diagram example (1/3)

- Business environment: College

- User requirements:
  - The college consists of several departments
  - Each department has its own head, and the head of the department must be a teacher
  - Teachers are assigned to departments
  - Teachers teach courses, and the course is always taught by only one teacher
  - Departments offer students multiple courses
  - Students enroll in courses

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ER diagram example (2/3)

- Entities and attributes (key underlined):
  - Institution (IDInstitution, Name, Address)
  - Department (IDDepartment, Name)
  - Teacher (IDTeacher, OIB, Name, Surname, Phone number)
  - Course (IDCourse, Name, ECTS)
  - Student (IDStudent, JMBAG, Name, Surname, Date of birth)
- Relations:
  - Consists, Has Leader, Assigned, Teaches, Offers, Enrolls
- Design decision:
  - There is no need for the Institution entity because the database is for one institution
- We draw an ER diagram and an entity diagram

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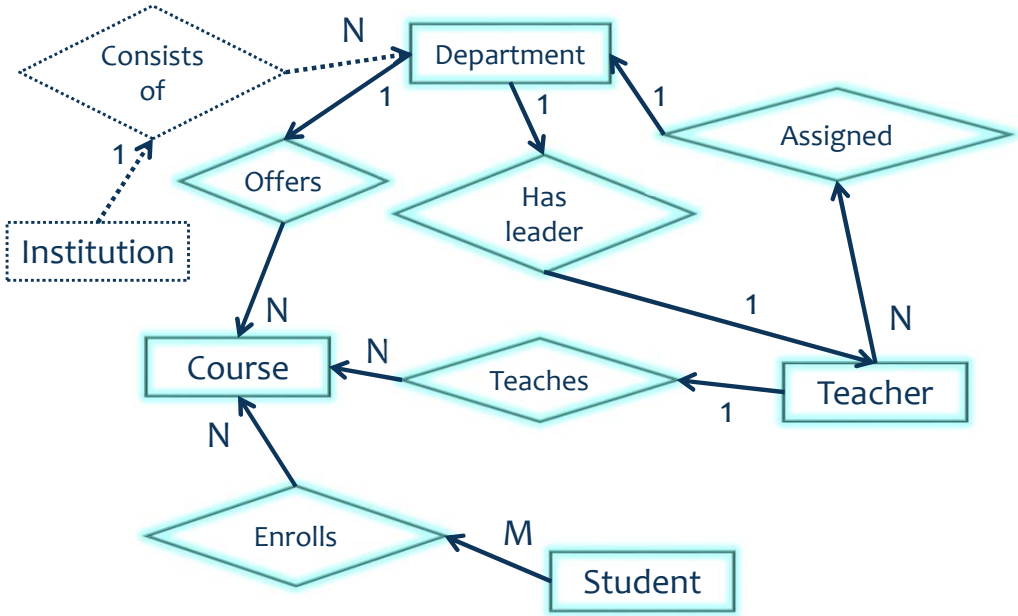
ER diagram example (3/3)



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Relational model

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Terminology

- There are several names for the same element of the relational model:

Mathematical name	Name in relational model	Traditional name
Relation	Table	File
Tuple	Row	Record
Attribute	Column	Field
Attribute value	Single data	Cell

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Table

- The basic elements of the relational model are interconnected tables
- Each table must have:
  - A name that is always in the singular
  - The structure made up of columns (also called a relational scheme)
  - Data made up of words

Table CountryRegion

	CountryRegionCode	Name
1	AD	Andorra
2	AE	United Arab Emirates
3	AF	Afghanistan
4	AG	Antigua and Barbuda
5	AI	Anguilla
6	AL	Albania
7	AM	Armenia
8	AN	Netherlands Antilles
9	AO	Angola

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Column domain

- The set of all values that a column can take
  - We say that the column is given on the domain
- Each column has exactly one domain
- Multiple columns can have the same domain
  - For example:
    - For the Hair Color column, the domain is a set of all possible colors
    - For the Price domain, the domain is the set of all real numbers
- Examples of multiple columns with the same domain:
  - Place of Birth and Place of Residence
  - Hair Color, Car Color and Font Color On Web Pages

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A row

- Columns define the structure
- **A row represents the data**
- Each row contains one value in each column
  - **If the column is optional, the row does not have to contain a value**
- Each row must be unique within the table
  - Must be different from all other rows in the table (key attribute)
- The set of all rows in a table makes up **the state of the table**
- The set of all rows in the database makes up **the state of the database**
- We say that the database can move from one state to another

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Thank You!

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